

Practical 2

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In [8]: import numpy as np
import pandas as pd
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split, cross_val_score, cross_val_predict, KFold
from sklearn.naive_bayes import GaussianNB
from sklearn.linear_model import LinearRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report, roc_auc_score, mean_squared_error, r2_score
from sklearn.preprocessing import label_binarize
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In [9]: # Load the Iris dataset
iris = load_iris()
X = iris.data
y = iris.target
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In [10]: # Split data into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
# Initialize and fit the Naïve Bayes classifier
nb_classifier = GaussianNB()
nb_classifier.fit(X_train, y_train)
# Predict on the test set
y_pred = nb_classifier.predict(X_test)
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In [11]: # Extract petal length and width for regression analysis
X_reg = X[:, 2].reshape(-1, 1) # Petal Length as feature
y_reg = X[:, 3] # Petal Width as target
# Split data into training and test sets
X_train_reg, X_test_reg, y_train_reg, y_test_reg = train_test_split(X_reg, y_reg, test_size=0.2, random_state=42)
# Initialize and fit the Linear Regression model
lin_reg = LinearRegression()
lin_reg.fit(X_train_reg, y_train_reg)
# Predict on the test set
y_pred_reg = lin_reg.predict(X_test_reg)
# Mean Squared Error
mse = mean_squared_error(y_test_reg, y_pred_reg)
# R2 Score
r2 = r2_score(y_test_reg, y_pred_reg)
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print(f"Mean Squared Error (MSE): {mse}")  
print(f"R2 Score: {r2}")
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Mean Squared Error (MSE): 0.045604284097661846

R2 Score: 0.9282562958836972

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In [12]: # 10-fold cross-validation for Linear Regression  
kfold = KFold(n_splits=10, random_state=42, shuffle=True)  
cv_mse = cross_val_score(lin_reg, X_reg, y_reg, cv=kfold, scoring="neg_mean_squared_error")  
cv_r2 = cross_val_score(lin_reg, X_reg, y_reg, cv=kfold, scoring="r2")  
print(f"Cross-Validation MSE: {-cv_mse.mean()}")  
print(f"Cross-Validation R2 Score: {cv_r2.mean()}")
```

Cross-Validation MSE: 0.04335630115789206

Cross-Validation R2 Score: 0.9203801852821047